

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY EXAMINATION

UCCC3073 / UCCG3073 DATA SCIENCE

FRIDAY, 20 MAY 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF SCIENCE (HONOURS)
STATISTICAL COMPUTING AND OPERATIONS RESEARCH

Instruction to Candidates:

This question paper consists of **FIVE (5)** questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in Section B, marks will only be awarded for the **FIRST (1)** question in that section in the order the candidate submits the answers.

Candidates are allowed to use any type of scientific calculator.

Answer questions only in the answer booklet provided.

SECTION A (Answer ALL questions)

- Q1. (a) Given the following customer records. Some records contain missing values and some contain typographical errors.

No.	First name	Last name	City	State
1	Lyndia	Tan	PJ	PJ
2	Tan	Lyndia	Petaling Jaya	Selangor
3	Jacynth		KL	
4	Dong	Tong	PJ	S'gor
5		Jenny		Shah Alam

Table Q1(a)

- (i) Provide TWO (2) possible reasons why such incompleteness occurs in the data. (4 marks)
 - (ii) Provide TWO (2) possible reasons why such inconsistency occurs in the data. (4 marks)
 - (iii) Record No. 1 and Record No. 2 look very similar in the data. Justify whether you will remove one of the records from the data. (4 marks)
 - (iv) Based on the answers in Q1(a)(ii) and Q1(a)(iii), suggest TWO (2) possible ways to reduce inconsistency in the data. (4 marks)
- (b) Consider the following data points:

92, 20, 10, 22, 35, 55, 110, 80, 15, 280, 240, 170

- (i) Given that the bin size is 3, partition these data points using the equal-width partition method. (4 marks)
- (ii) Smooth the bins in Q1(b)(i) using bin boundaries. (3 marks)
- (iii) Provide ONE (1) advantage and ONE (1) disadvantage of data smoothing. (2 marks)

[Total : 25 marks]

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- Q2. Given the following training data set. The data contain missing values in the Type attribute.

App ID	Price (RM)	Type	Review	Buy?
1	2.99	Game	Fair	No
2	2.50	Productivity	Good	No
3	3.49	Camera	Fair	Yes
4	1.99	Game	Fair	Yes
5	4.99	Productivity	Average	Yes
6	5.90	Game	Average	No
7	3.20	Game	Good	Yes
8	3.50	-	Good	No
9	4.20	Camera	Average	Yes
10	3.49	Camera	Good	Yes

Table Q2

The test instance is given as follow:

App ID	Price	Type	Review	Buy?
11	4.60	Camera	Good	?

A Bayesian model is used to train and test the data.

- State any possible errors that might occur when calculating probability for the test data. (2 marks)
- Suggest a solution to overcome the error stated in Q2(a). (2 marks)
- Assume that data imputation is not applied to the training data. Using the solution proposed in Q2(b), calculate the probabilities for attributes Type, Review and Buy. Calculate the mean and standard deviation for attribute Price. Round your answer to the nearest 2 decimal points. (8 marks)
- Given that the probability for attribute Price in the Yes group is 0.8484 and in the No group is 0.72, calculate the likelihood of customer App ID 11 to purchase the camera. Show the working steps in your answer. Round your answer to the nearest 2 decimal points. (5 marks)
- Assume that data imputation is applied to the training data. Determine the imputed value and calculate all possible probabilities for the Type attribute. (4 marks)

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- (f) Using the imputed value in Q2(e), justify whether the likelihood of the customer App ID 11 purchase the camera is higher or lower than the likelihood computed with no data imputation. (4 marks)

[Total : 25 marks]

- Q3. Assuming K-nearest neighbour (KNN) classifier is used. Table Q3 shows a training set of positive (+) and negative (-) instances.

x	3.0	4.5	4.6	5.2	5.3	5.5
y	-	+	+	-	-	+

Table Q3

You would like to classify the test instance z ($z = 5.0$) to the nearest class with the following K neighbours:

$K = 1$

$K = 3$

$K = 5$

- (a) Using the majority vote approach, classify the test instance z . Show the working steps in your answer. (10 marks)
- (b) Using the distance-weighted approach, classify the test instance z . show the working steps in your answer. (10 marks)
- (c) Based on your classification results on Q3(b) and Q3(c), justify whether you would recommend setting $K = 3$ for both majority vote and distance-weighted. (3 marks)
- (d) Provide ONE (1) disadvantage of KNN. (2 marks)

[Total : 25 marks]

UCCC3073 / UCCG3073 DATA SCIENCE**SECTION B (Choose only ONE question)**

- Q4. You are hired as a data scientist by a company. You are given the following confusion matrix for two classification models and are required to propose the best model to the company.

N = 1,000		Prediction values			
		Model 1		Model 2	
		No	Yes	No	Yes
Actual values	No	100	0	90	10
	Yes	700	200	500	400

Table Q4

- (a) Calculate the baseline performance. (2 marks)
- (b) Calculate the following evaluation measures for both models. For each measure, identify the best model.
- (i) Classification accuracy rate. (3 marks)
 - (ii) Precision. (3 marks)
 - (iii) Specificity. (3 marks)
 - (iv) F₁ score. (3 marks)
- (c) Using the computed results in Q4(a) and Q4(b), justify whether you would propose the model to the company. (5 marks)
- (d) Suggest TWO (2) ways to overcome issues you have observed in Q4(c). (6 marks)

[Total : 25 marks]

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- Q5. UTAR is conducting a study survey on student response to attend classes physically. Decision tree is used to predict student responses. Category 1 indicates positive response (i.e. happy to return to campus) and category 0 indicates negative response (i.e. reluctant to return to campus).

- (a) Table Q5(a) is the confusion table of the decision tree model:

N = 1,000		Decision tree prediction with equal cost	
		0	1
Actual	0	101	199
values	1	220	480

Table Q5(a)

- (i) Calculate the baseline performance. (2 marks)
- (ii) Calculate misclassification rate, sensitivity, specificity and F_1 score. (4 marks)
- (iii) Justify whether or not the model is fit for the data. (5 marks)
- (b) Assume that unequal cost on false positive (FP) and true positive (TP) are implemented in the decision tree model, the following is the confusion table of the model when cost for false positive, $Cost_{FP}$ is 10 and cost for true positive, $Cost_{TP}$ is -40.

N = 1,000		Decision tree prediction with unequal cost	
		0	1
Actual	0	140	160
values	1	50	650

Table Q5(b)

- (i) Calculate misclassification rate, sensitivity, specificity and F_1 measure. (4 marks)
- (ii) Calculate the model profit per record. (3 marks)
- (iii) Justify whether or not the unequal cost model is better than the baseline performance and the equal cost model in Q5(a). (7 marks)

[Total : 25 marks]

APPENDIX

Regression	
Straight line, $\hat{y} = \beta_0 + \beta_1 x$	t value, $t = \frac{\hat{\beta}}{SE}$
Coefficients $\hat{\beta}_1 = \frac{p \sum xy - \sum x \sum y}{p \sum x^2 - (\sum x)^2}$	Standard errors of coefficients $SE(\hat{\beta}_1) = \frac{s}{\sqrt{\sum (x - \bar{x})^2}}$
$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$,	$SE(\hat{\beta}_0) = s \sqrt{\frac{1}{p} + \frac{\bar{x}^2}{\sum (x - \bar{x})^2}}$
Coefficient of Determination $R^2 = \left[\frac{p \sum xy - \sum x \sum y}{\sqrt{p \sum x^2 - (\sum x)^2} * \sqrt{p \sum y^2 - (\sum y)^2}} \right]^2$	Residuals $\hat{\varepsilon} = y - \hat{y}$
$R = \sqrt{R^2}$	$s^2 = \frac{\sum \hat{\varepsilon}^2}{p - 2}$
	$s = \sqrt{s^2}$

Naïve Bayes
Bayes' rule, $P(H E) = \frac{P(E H)P(H)}{P(E)}$
Laplace correction, $\frac{x_i+1}{c+\alpha}$
Probability density function for a Gaussian distribution, $pdf(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\bar{x})^2}{2\sigma^2}}$

Association Rules
Support, $s = \frac{freq(X,Y)}{N}$
Confidence, $c = \frac{freq(X,Y)}{freq(X)}$
Lift, $l = \frac{Support}{Supp(X)*Supp(Y)}$

Evaluation metrics	
Classification accuracy, $acc = \frac{TN+TP}{TN+TP+FN+FP}$	Precision, $PPV = \frac{TP}{TP+FP} = 1 - FDR$
Misclassification rate, $1 - acc$	F1 score, $F_1 = \frac{2TP}{2TP+FP+FN}$
Sensitivity / Recall, $recall = \frac{TP}{TP+FN}$	Overall model cost $= TN.Cost_{TN} + FP.Cost_{FP} + FN.Cost_{FN} + TP.Cost_{TP}$
Specificity, $spec = \frac{TN}{TN+FP}$	Model cost per case $= \frac{\text{Overall model cost}}{N}$
False discovery rate, $FDR = \frac{FP}{FP+TP}$	Model profit per case $= -\text{model cost per case}$

Proximity measures	
Euclidean distance, $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$	Hamming distance, $d(a, b) = (a \oplus b)$
Manhattan distance, $d(x, y) = \sum_{i=1}^n x_i - y_i $	Simple matching coefficient, $SMC = \frac{\text{num of matching attribute values}}{\text{num of attributes}}$
Supremum distance, $d(x, y) = \max(x_i - y_i)$	Jaccard coefficient, $J = \frac{\text{num of matching presences}}{\text{num of attributes not involved in 00 matches}}$
	Cosine similarity, $\cos(x, y) = \frac{x \cdot y}{ x y }$

Miscellaneous	
Min-Max normalisation, $x' = \frac{x - \min(x)}{\max(x) - \min(x)}$	Confidence intervals, $CI = \bar{x} \pm z * SE$
z-score normalisation, $x' = \frac{x - \mu}{\sigma}$	Standard error, $SE = \frac{\sigma}{\sqrt{n}}$
Sample mean, $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	Z-test, $Z = \frac{\bar{x} - \mu}{SE_{\bar{x}}}$
Sample standard deviation, $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$	